

Supplementary Methods

Method for the two-stage likelihood ratio test (2sLRT)

To analyze unreplicated RNA-seq data, we developed a two-stage likelihood ratio test (2sLRT) to identify DMP (deviation from mid-parent) genes across multiple families (Qi and Liu, 2025). This test is based on the negative binomial distribution, which is widely recognized as an appropriate statistical model for RNA-sequencing count data. The key idea of our method is to borrow information across multiple families to estimate dispersion parameters, which are necessary for negative binomial models.

The first stage estimates dispersion parameters through a three-step process. In the first step, we apply a Poisson-based likelihood ratio test for differential expression analysis with False Discovery Rate (FDR) control to identify "null families" for each gene. A null family is a family where there is no differential expression across the two parental lines and the hybrid, so that the three genotypes can be viewed as "pseudo-replicates." In the second step, we keep genes with a sufficient number of null families (more than N_0) and cluster genes according to their patterns of null families using Jaccard distance computed from the indicator matrix of each family being "null" or not. This ensures genes with similar null-family profiles are grouped together. Within each cluster, we identify the gene with the minimum number of null families and use its null families as "pseudo-treatments", each with three pseudo-replicates, to estimate the dispersion parameter for genes in that cluster using the well-established edgeR method. In the third step, we fit a Locally Estimated Scatterplot Smoothing (LOESS, Cleveland, 1979) curve, that models the relationship between these calculated dispersion values and median DMP expression levels and use this curve to get dispersion parameter estimates for genes not included in the second step.

In the second stage of the 2sLRT method, we use dispersion parameters estimated from the first stage to conduct a negative binomial likelihood-ratio test and obtain a p-value for testing of DMP for each gene in each family. The mathematical derivation of this framework is detailed in Qi and Liu (2025).

Applying the 2sLRT algorithm requires several tuning parameters. For our dataset with 19880 genes and 599 families, the tuning parameters are as follows: an FDR threshold (α_0) of 0.1 for identifying null families with Poisson LRT test, a minimum family count (N_0) of 50 to be included in the clustering step of stage 1, a cluster count (K) of 10, and a LOESS span of 0.1.

Validation procedures for the 2sLRT method

We validated our method using both simulations and independent replicated RNA-Seq datasets. We first evaluated the control of FDR; this is to ensure results are reliable without many false positives. This evaluation requires the knowledge of the ground truth about DMP genes, so we checked FDR control using the house-keeping genes and simulated datasets. We also evaluated the ranking of DMP genes and compare the ranking obtained from unreplicated data to the replicated data using the Area Under the ROC Curve (AUC). Higher values of AUC indicate

agreement of rankings between results from unreplicated and replicated data. This evaluation is applicable to real RNA-Seq datasets without ground truth.

To evaluate the FDR control of the 2sLRT, we first applied our method to the known house-keeping genes where DMP is not expected. We reported the percentage of significant families for each house-keeping gene. This percentage should be very low, indicating our method successfully controls the false positives. As shown in the results, the proportion of the corresponding gene being identified as DMP across 599 families when FDR was controlled at 5% (Table 3 in Qi and Liu, 2025). It clearly shows that a Poisson test for DMP (that can be applied to unreplicated count data) generates many false positive claims, while the 2sLRT controlled the proportion of false positives well below 0.05.

Second, we conduct a simulation study comparing our method against a standard approach using three replicates. We compare the FDR computed from our method to the replicated results (Table S4). Across all scenarios (varying gene counts, number of families, and variation levels), our method maintained strict FDR control, consistently keeping the actual FDR below the controlled level (5%). This confirms that our method is statistically conservative and does not yield many false positives. As a plus, we also compare the detection power. Although replicated designs yield near-perfect statistical power, our method achieves respectable power that scales with dataset size, improving from 50% to 70% as the number of families increases from 100 to 200. Ultimately, by prioritizing family count over replicates, we trade the power to detect (possibly) weak effects in specific families for the ability to map DMP across much broader genetic diversity.

Comparison of 2sLRT results with real-world replicated datasets

We further compared results for unreplicated data based on our 2sLRT method and methods based on replicates using two real RNA-seq datasets that have biological replicates by comparing the ranking of DMP genes obtained from unreplicated data to the replicated data using AUC.

The first dataset was retrieved from Pitz et al., 2025. This study utilized an alpha-design with incomplete blocks to identify heterosis genes in maize. The experiment has three complete biological replicates, within which genotypes were distributed across multiple batches over time to accommodate the large sample size (~1,152 samples). The genetic material includes 112 Recombinant Inbred Lines (RILs) and their corresponding testcross hybrids (RIL $\times$ B73 and RIL $\times$ Mo17). To effectively correct for batch effects and spatial variation within the aeroponic growth systems, the parental inbred lines (B73 and Mo17) were embedded as common checks across all batches and systems.

Prior to method validation, we assessed the magnitude of technical variation using the parental inbred lines (B73 and Mo17), which were highly replicated across the experiment. We performed a Variance Component Analysis on the highly replicated parental lines, B73 and Mo17, using a linear mixed-effects model with random effects for replicate, batch, and system. The analysis revealed that System effects were a major source of technical noise, explaining a median of 41.1% of the variance in B73 and 13.7% in Mo17. Batch effects accounted for an additional 6.4-9.0% of the variance. In contrast, biological replicate effects contributed

minimally ($< 2\%$). These results demonstrate that batch and system technical factors introduce substantial variability. These cannot be ignored in the single replicate analysis.

To validate our single-replicate method without confounding technical variation, we restricted our analysis to families where the hybrid and both parents were processed in the same experimental batch and grown in the exact same aeroponic system. We identified a total of 169 such families available for validation (39 in Replicate 1, 65 in Replicate 2, and 65 in Replicate 3). All of these families were tested but note that since $B73 \times Mo17$ and their parental genotypes were highly replicated, some of the identified families have the same genotype, i.e., $B73 \times Mo17$. Thus, in the final comparison with the replicated analysis, we used only the best-performing family for this hybrid.

For each independent replicate run, we treated the data as if it were the only available samples and subset the raw count matrix to include only the samples corresponding to the identified matched families. A pseudo-count of 1 was added to all raw counts to ensure numerical stability in downstream algorithms. We then performed standard preprocessing as in the original paper using the `edgeR` package (Robinson et al., 2010): raw counts were normalized using the TMM method to account for sequencing depth, and lowly expressed genes were filtered using the `filterByExpr` function based on Genotype grouping. We applied our proposed method 2sLRT to the families we identified in this replicate. For the dispersion estimation stage, parameters were optimized per replicate: for Replicate 1 ($N_0 = 15, \alpha_0 = 0.1, K = 1$); for Replicate 2 ($N_0 = 20, \alpha_0 = 0.01, K = 1$); and for Replicate 3 ($N_0 = 12, \alpha_0 = 0.01, K = 1$), where N_0 denotes the minimum number of null families required for inclusion, α_0 the significance threshold for the Poisson null-family test, and K the number of gene clusters. We then compared the significant gene-family combinations from each single-replicate run against the results from using the replicated data. P-values were adjusted using the Benjamini-Hochberg procedure across all genes in each family at a nominal level of 0.05.

In comparison, we analyzed the full replicated dataset using a rigorous Generalized Linear Mixed Model (GLMM) approach. For fair comparison and computational efficiency, when comparing the results from 2sLRT applied to each replicate, this GLMM approach is applied only to the full families in this replicate. Specifically, raw counts were normalized using the TMM method. Genes with low expression were filtered using a design matrix accounting for Genotype to ensure detectability. Robust tagwise dispersion parameters were estimated using `edgeR` with the design $\sim$ Genotype + Batch, utilizing all available replicates to characterize biological noise. We implemented a GLMM using `glmmTMB` with a Negative Binomial distribution (`nbinom2`). Then we utilized a log-link function but performed hypothesis testing on the response scale. The model included fixed effects for Genotype and Batch, and a random intercept for System. Crucially, the dispersion parameter for each gene was fixed to the robust `edgeR` estimate to prevent convergence failures due to the limited sample size of specific genotypes. Marginal means were estimated and back-transformed to the arithmetic scale using the `emmeans` package. Significance was determined for the contrast $\mu_{Hybrid} - 0.5(\mu_{P1} + \mu_{P2}) = 0$ using the Delta Method to approximate standard errors. P-values were adjusted using the Benjamini-Hochberg (FDR) procedure across all genes in each family.

For each of the three replicates, we treated the results from replicated data analysis at the FDR level 0.05 as the benchmark and computed the AUC to compare the unreplicated result with the benchmark.

The second dataset is an in-house dataset comprising 48 samples, representing 10 hybrids and their parental genotypes with two biological replicates. Data preprocessing included normalization using DESeq2 (Love et al., 2014) and the removal of genes containing zero counts in any sample, resulting in a final set of 24,241 genes. We first applied the replicated negative binomial method to the full dataset to establish a benchmark.

To evaluate the performance of the 2sLRT method, we employed a random subsampling strategy repeated 10 times. In each iteration, the dataset was partitioned into two independent sets, ensuring that each set contained exactly one replicate per genotype. We then applied the 2sLRT method to these individual replicates. For dispersion estimation, we used the parameters $N_0 = 2$, $\alpha_0 = 0.01$, and $K = 3$ (where N_0 is the minimum null families, α_0 is the Poisson null-family test threshold, and K is the number of gene clusters). The performance is then evaluated against the benchmark results obtained from replicated data analysis at FDR level of 0.05.

Finally, to evaluate significance for each gene across all families, we applied a one-sided Kolmogorov-Smirnov (KS) test (Massey Jr, 1951). This test is to identify genes whose distribution of p-values is stochastically smaller than Uniform (0,1), corresponding to genes that tend to be non-additive across families. To ensure strict statistical validity, the Benjamini-Hochberg procedure (Benjamini and Hochberg, 1995) was applied to these KS test results, effectively controlling the final overall FDR at 0.05. The proportion of all hybrids for a gene to be detected non-additive at an FDR of 0.05 was also calculated, and this proportion is at most 5.18% among genes with an insignificant KS test. We declared a gene tends to be non-additive based on both FDR control of the KS test and the threshold of minimal proportion of non-additive families (5.18%).

References

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